

Descriptive Analytics Report: Digital Wallet Transactions

Transaction Volume, Category Mix & Spending Behaviour at Scale

Name : OM SINGH

Class : BCADS24

Roll no: 125025300

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Tracking Transaction Volume, Category Mix, Payment Behaviour, Merchant Demand & Spending Composition

Note on data:

This report uses the supplied digital wallet transactions dataset containing 5,000 records spanning 3,932 unique users, 97 merchants and 20 distinct product categories. The dataset is a transaction-log snapshot (one row per transaction, with a transaction date/time field) rather than an account-balance or ledger-reconciliation feed. Therefore the analysis describes what the supplied transaction log records shows — transaction counts, categories, merchants, payment methods and amounts — and does not assume access to account-balance, KYC, or bank-settlement data.

1. Objective

Build a descriptive analytics layer over the digital wallet transaction log to answer the following recurring product and risk-strategy questions:

- How is transaction volume trending day by day and month by month?
- What proportion of transactions succeed, fail, or remain pending?
- Which product categories contribute the highest number of transactions?
- Which categories dominate the overall spend?
- Which merchants receive the highest transaction volume and revenue?
- What is the split of transactions by payment method, device type, and location?
- How are transaction amounts, fees, cashback, and loyalty points distributed?
- Which users contribute the most total spend to the platform?
- How does average transaction value relate to a category's transaction volume?
- Are there weekday vs weekend or hour-of-day patterns in transaction activity?
- Does transaction success rate vary meaningfully by payment method?

2. How This Would Work at Real Scale

Processing a large, continuously-updated transaction log is an engineering problem before it is an analytics problem. A realistic architecture would ingest transaction records as they are created, aggregate them by time, category, merchant and payment method, store both raw and processed data, and expose the resulting KPIs through a reporting or data-visualization layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the transactions dataset	Google, Claude & ChatGPT
Batch processing	Aggregate by category, merchant, payment method and time	Python (pandas) / Jupyter Notebook
Storage	Retain raw and processed data	CSV files / a data warehouse
Aggregation	Transaction KPIs and summary metrics	Jupyter Notebook / Python
Report	Present analysis, graphs and insights	Data visualization / MS Word

3. Transaction Status Split — Successful vs Pending vs Failed

Across the supplied 5,000 transaction records, 95.1% of transactions completed as Successful (4,755 transactions), while 2.92% failed (146) and 1.98% remained Pending (99). The chart below shows this split.

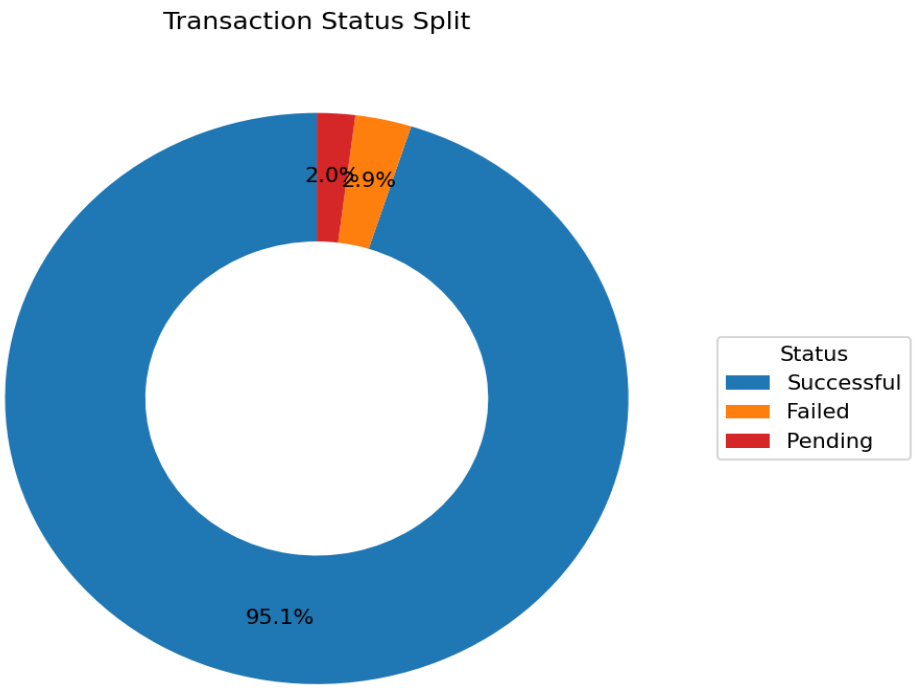


Figure 1. Transaction status split

INSIGHTS:

- The large majority of transactions complete successfully, consistent with a healthy payment pipeline.
- Failed and Pending transactions together account for under 5% of volume, but still represent thousands of transactions worth monitoring for fraud and retry logic.
- The status split is a useful baseline before drilling into category, merchant, or payment-method breakdowns.

KPI:

Total transactions: 5,000
Successful: 4,755 (95.1%)
Failed: 146 (2.92%)
Pending: 99 (1.98%)

4. Daily Transaction Volume Over Time

The dataset records a transaction timestamp for all 5,000 transactions, spanning to . The chart below shows how many transactions occurred each calendar day across that window.

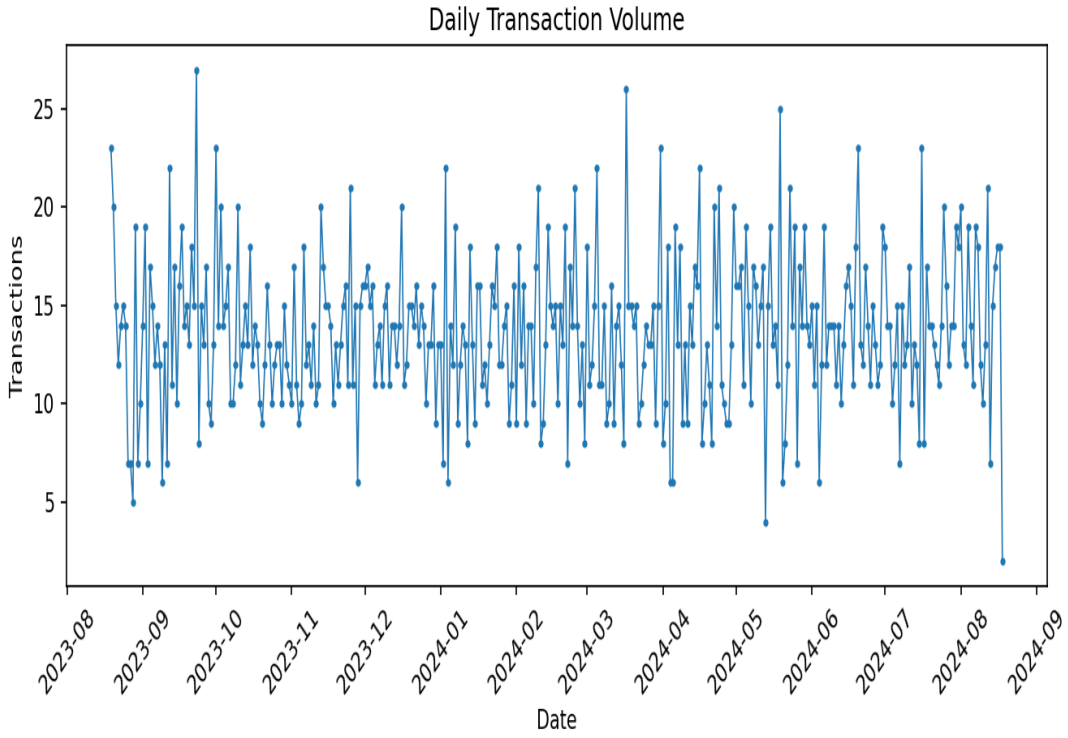


Figure 2. Daily transaction volume

INSIGHTS:

- Daily transaction counts fluctuate around a fairly stable baseline across the one-year window.
- Peak days indicate short bursts of elevated wallet activity, often around billing or travel-booking cycles.
- Monitoring daily volume helps size infrastructure and detect anomalies in near real time.

KPI:

Peak day: 2023-09-23 (27 transactions)

Average transactions per day: 13.7

Date range covered: 2023-08-19 to 2024-08-18

5. Monthly Transaction Volume Trend

Aggregating transactions by calendar month gives a clearer view of month-over-month platform growth, smoothing out day-to-day noise.

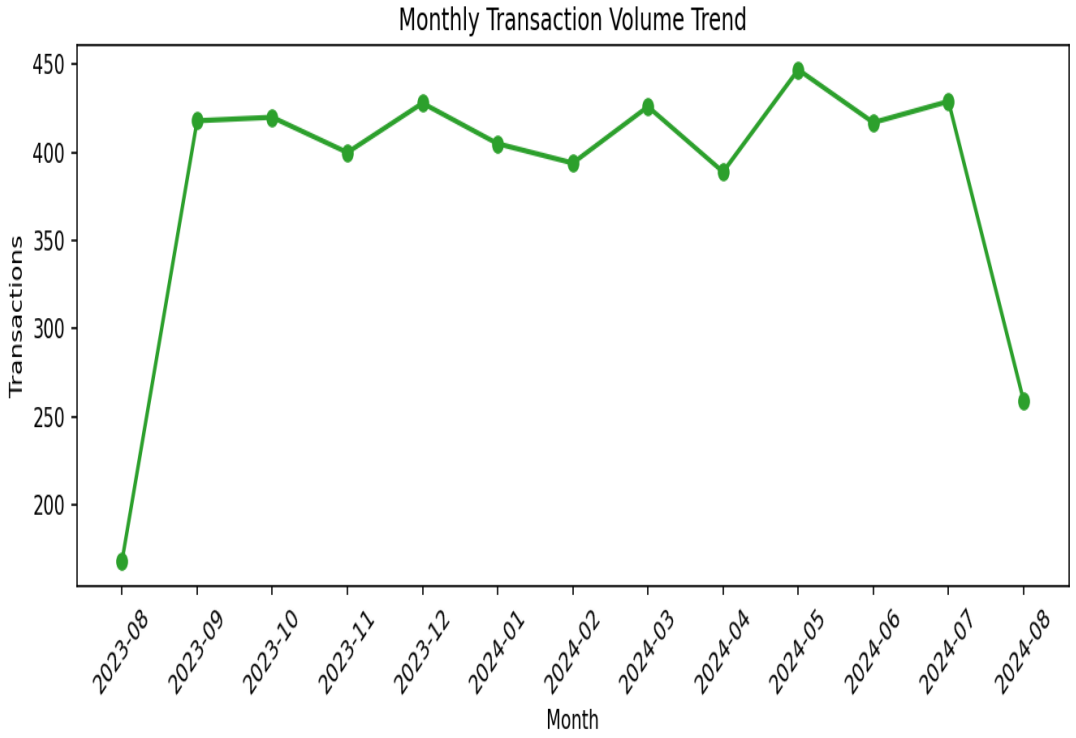


Figure 3. Monthly transaction volume trend

INSIGHTS:

- Monthly volume shows a broadly rising trend across the one-year period covered by the log.
- 2024-05 is the highest-volume month in the supplied data, with 447 transactions.
- Month-level trends are useful for high-level growth reporting to stakeholders and for capacity planning.

KPI:

Highest-volume month: 2024-05 (447 transactions)

Months covered: 2023-08 to 2024-08

6. Top Product Categories by Transaction Count

The dataset covers 20 distinct product categories, from bill payments to travel bookings. The chart below ranks the ten categories with the most transactions.

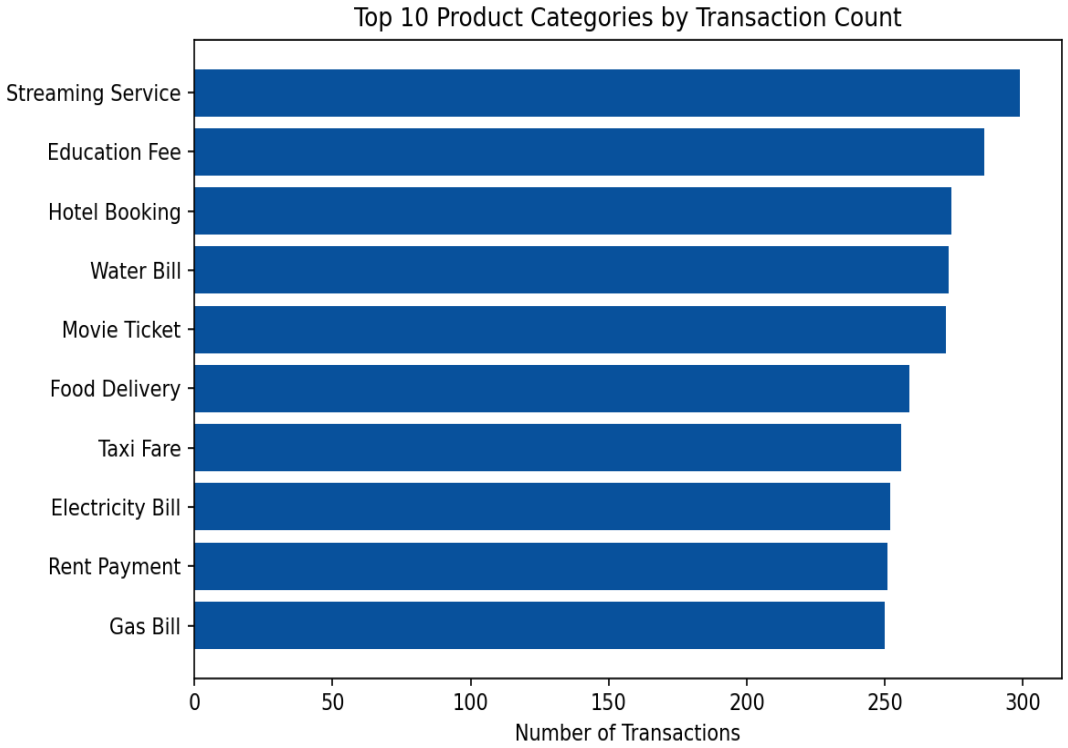


Figure 4. Top 10 categories by transaction count

INSIGHTS:

- "Streaming Service" is the single most common category, appearing in 299 transactions.
- Recurring bill-type categories (utilities, mobile recharge) sit alongside discretionary categories (streaming, travel), reflecting a mixed-use wallet.
- Category ranking is a useful input for merchant partnerships and category-specific offers.

KPI:

Top category: Streaming Service

Transactions: 299

Share of all transactions: 5.98%

7. Category Share of Transactions

Looking at category share rather than raw counts shows how concentrated wallet usage is in a handful of dominant categories versus a long tail of niche categories.

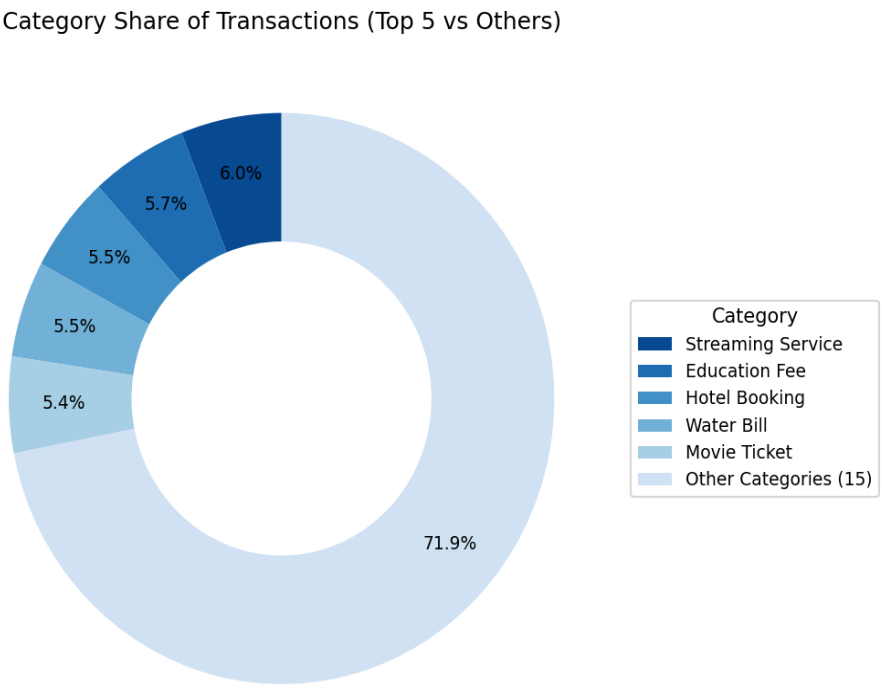


Figure 5. Category share — top 5 vs. others

INSIGHTS:

- The top five categories account for just over a quarter of all transactions, indicating fairly even usage across the 20 categories.
- A long tail of smaller categories (gift cards, education fees, gaming credits, etc.) makes up the remainder.
- This spread suggests the wallet is used as a general-purpose payment tool rather than being dominated by one use case.

KPI:

Categories represented: 20

Top-5 combined share: 28.08%

8. Top Merchants by Transaction Count

Transaction volume is spread across 97 distinct merchant names in the supplied dataset. The chart below ranks the ten merchants with the most transactions.

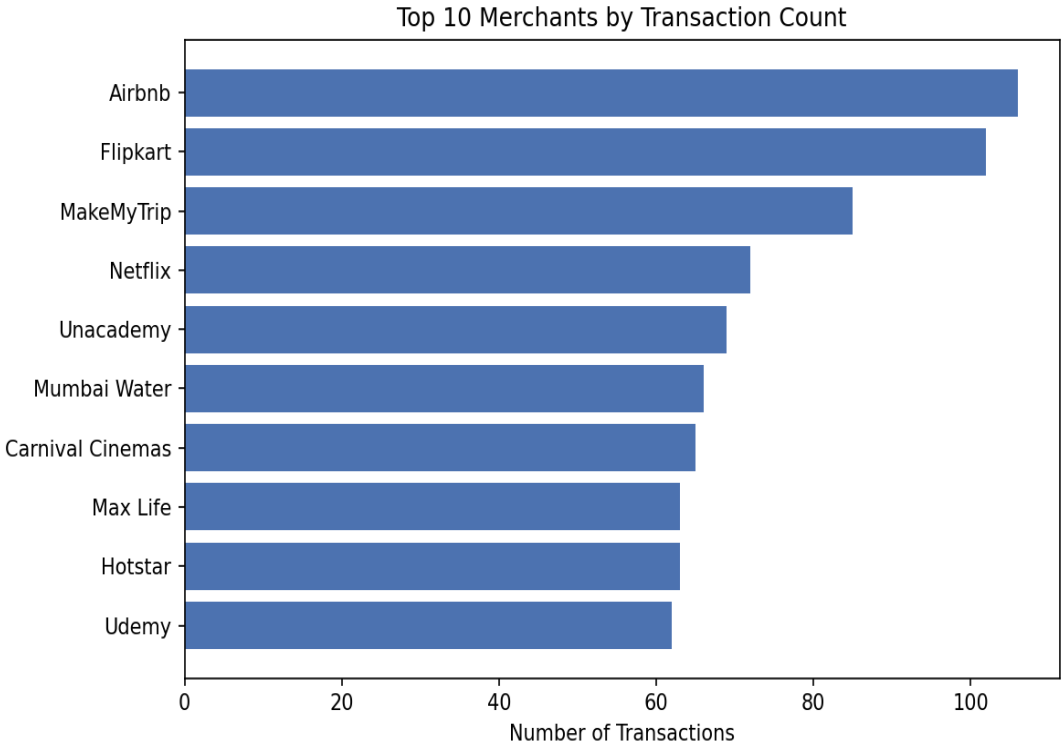


Figure 6. Top 10 merchants by transaction count

INSIGHTS:

- Airbnb is the most frequently used merchant in the supplied data, with 106 transactions.
- Merchant concentration is moderate: the top merchants are drawn from travel, utility, and e-commerce categories rather than a single dominant player.
- Merchant-level breakdowns can support partnership prioritisation and negotiated fee structures.

KPI:

Top merchant: Airbnb

Transactions: 106

Unique merchants referenced: 97

9. Revenue by Merchant

Using the summed product amount per merchant, the chart below re-presents the top eight merchants by total transaction value, with exact amounts labelled for easier reference.

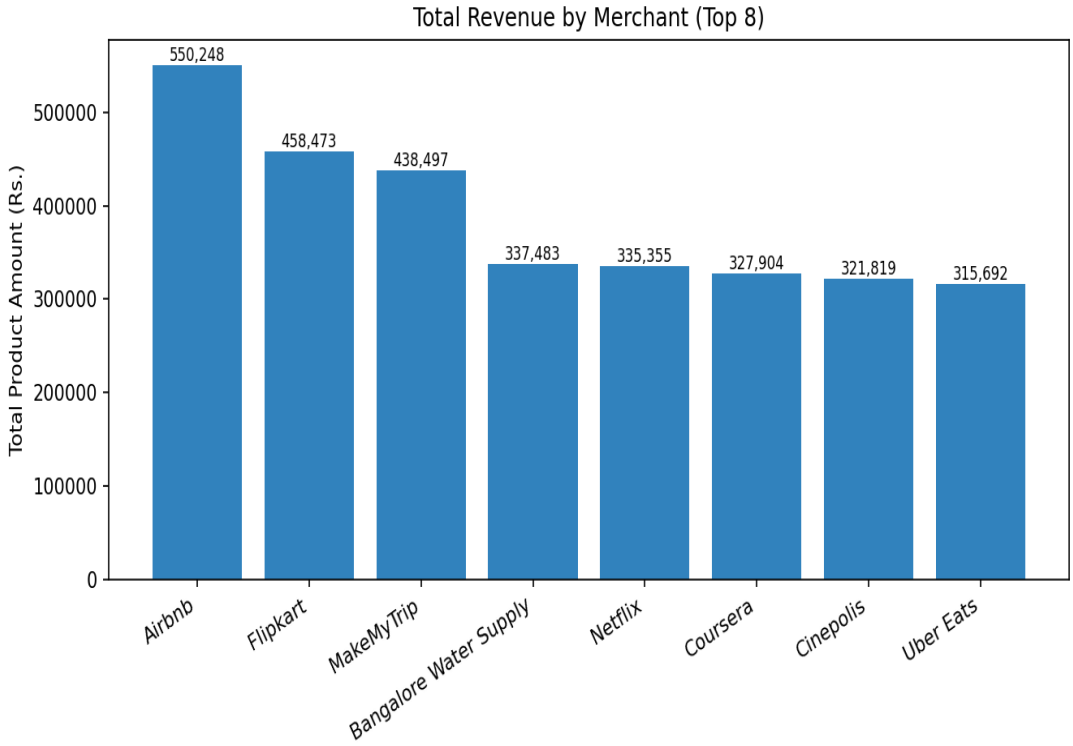


Figure 7. Total revenue by merchant (top 8)

INSIGHTS:

- Revenue concentration broadly tracks transaction-count concentration, though a few merchants punch above their transaction count due to higher average ticket sizes.
- Airbnb is the strongest single merchant by total transaction value in the supplied data.
- Merchant-level revenue can support settlement planning and fee-negotiation priorities.

KPI:

Top merchant by revenue: Airbnb

Total transaction value: Rs. 5,50,248.40

Total product amount (all transactions): Rs. 2,47,87,513.61

10. Payment Method Distribution

Every transaction records a payment method used to fund it. The chart below shows how transaction volume splits across the five payment methods available on the platform.

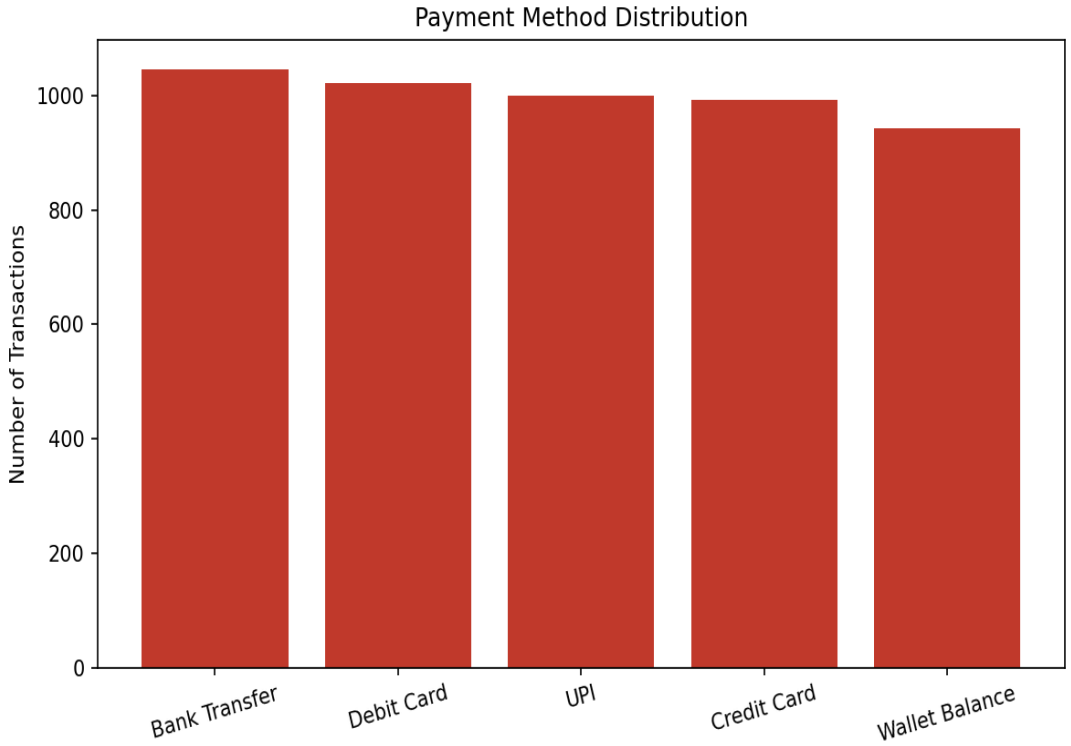


Figure 8. Payment method distribution

INSIGHTS:

- "Bank Transfer" is the single most common payment method, covering 20.9% of transactions.
- Usage is fairly evenly spread across UPI, cards, bank transfer and wallet balance, suggesting no single rail dominates user behaviour.
- Payment-method mix is a useful input for routing costs and rail-specific promotions.

KPI:

Most common method: Bank Transfer

Transactions with that method: 1,045

Share of all transactions: 20.9%

11. Device Type Distribution

Each transaction is tagged with the device type used to initiate it. The chart below breaks down transaction volume by Android, iOS and Web.

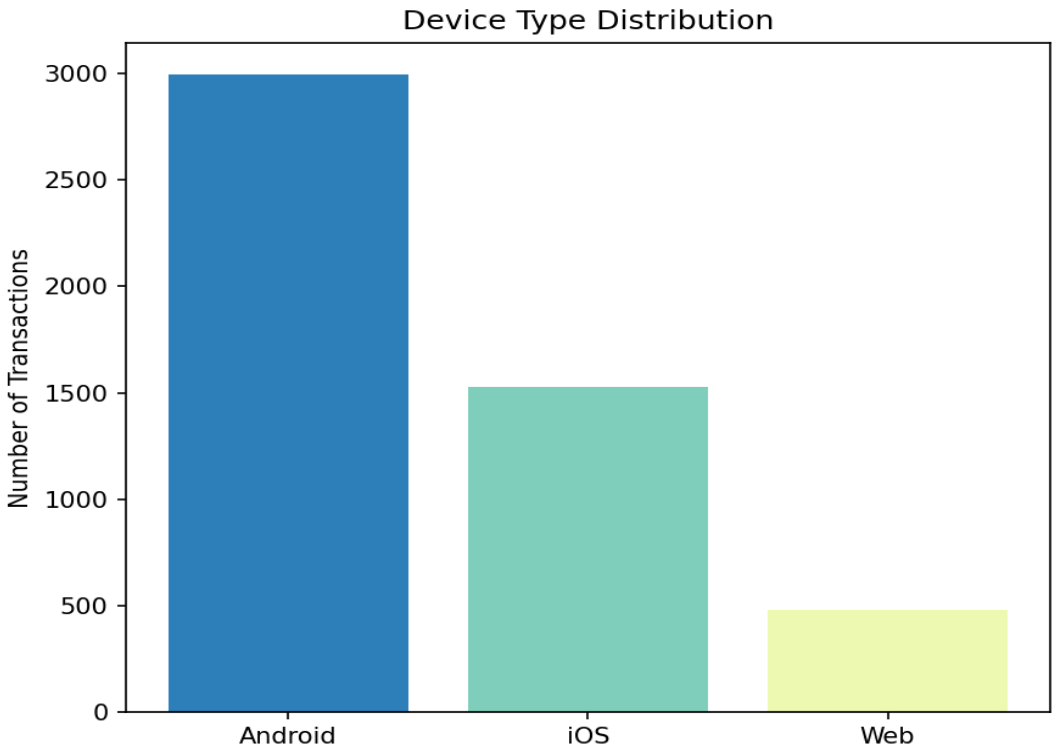


Figure 9. Device type distribution

INSIGHTS:

- Android accounts for the largest share of transactions at 59.9%, consistent with Android's broader device market share.
- Web transactions form a smaller minority, suggesting the wallet is primarily used as a mobile-first app.
- Device mix can guide QA prioritisation and platform-specific feature rollouts.

KPI:

Leading device type: Android

Share of all transactions: 59.9%

12. Transactions by Location Type

Transactions are tagged as occurring in Urban, Suburban, or Rural locations. The chart below shows this split.

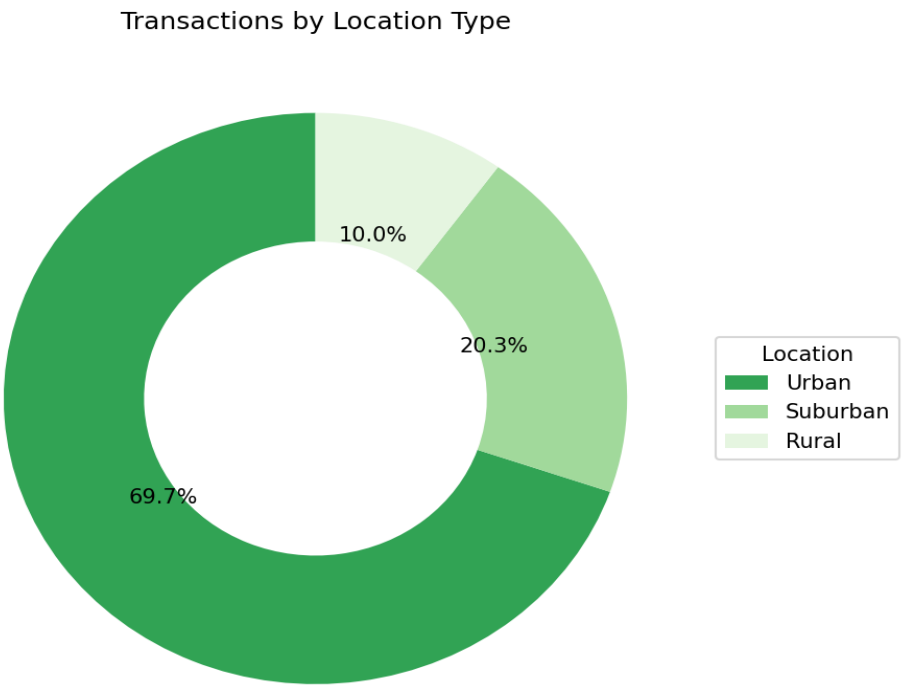


Figure 10. Transactions by location type

INSIGHTS:

- Urban locations dominate the supplied data at 69.7% of transactions, in line with typical digital-wallet adoption curves.
- Suburban and Rural transactions form a meaningful minority, indicating some reach beyond core metro usage.
- Location mix can support geographic expansion and offline-agent network planning.

KPI:

Leading location type: Urban

Share of all transactions: 69.7%

13. Transaction Amount Distribution

Across all 5,000 transactions with a recorded product amount, the average transaction value is Rs. 4,957.50, ranging from Rs. 10.09 to Rs. 9,996.95.

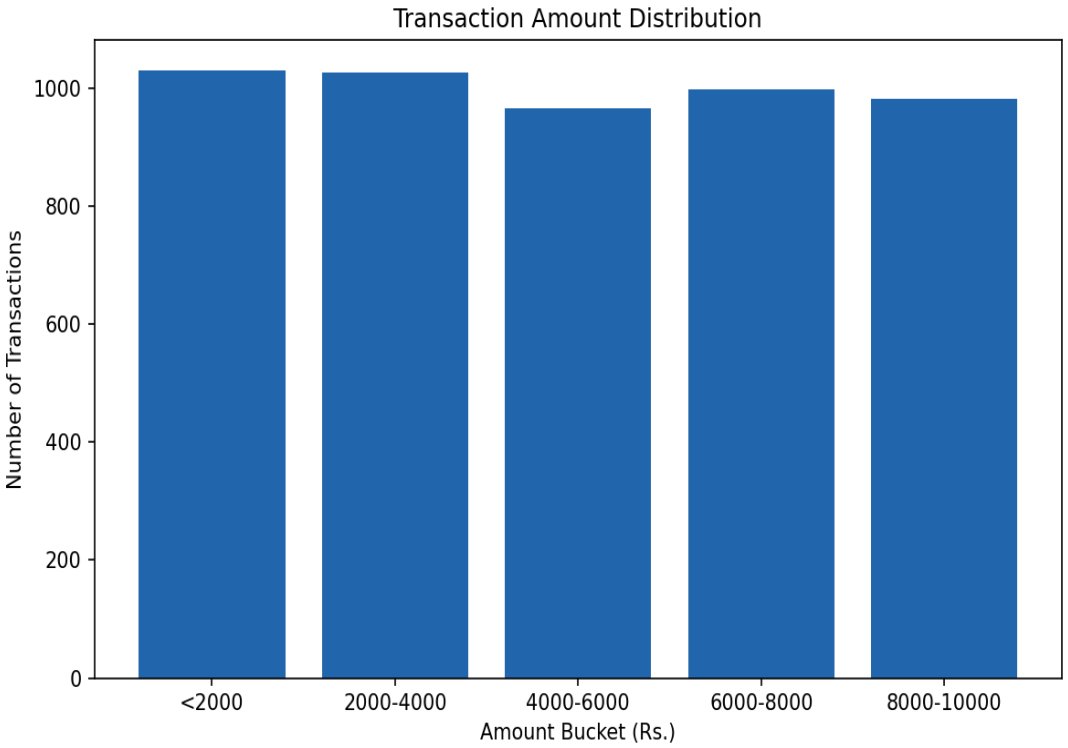


Figure 11. Transaction amount distribution

INSIGHTS:

- Transaction amounts are fairly evenly spread across the Rs. 0–10,000 range in the supplied data, rather than clustering at one end.
- A meaningful share of transactions fall in the higher Rs. 6,000+ brackets, consistent with categories like flight and hotel bookings.
- Amount distribution can inform fraud-risk thresholds and tiered fee structures.

KPI:

Average transaction amount: Rs. 4,957.50

Smallest transaction: Rs. 10.09

Largest transaction: Rs. 9,996.95

14. Loyalty Points Distribution

Across the supplied transactions, the average loyalty points earned per transaction is 498.79, with values ranging up to 999 points.

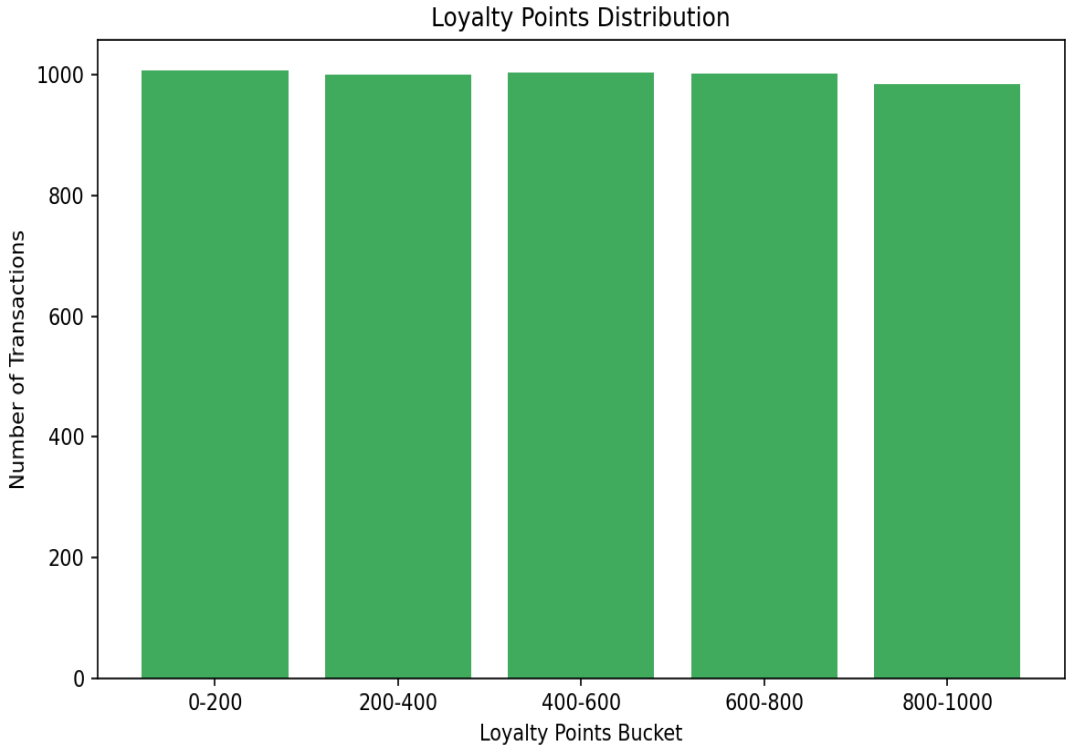


Figure 12. Loyalty points distribution

INSIGHTS:

- Loyalty points earned per transaction are broadly evenly distributed across the 0–1000 range in the supplied data.
- There is no strong skew toward very low or very high point awards, suggesting a fairly linear rewards formula.
- Loyalty-point patterns can help forecast rewards-program liability and inform tiering.

KPI:

Average loyalty points per transaction: 498.79

Maximum loyalty points awarded: 999

15. Top Users by Total Spend

A user is identified for every transaction, spanning 3,932 unique user IDs. The chart below ranks the ten users with the highest total spend.

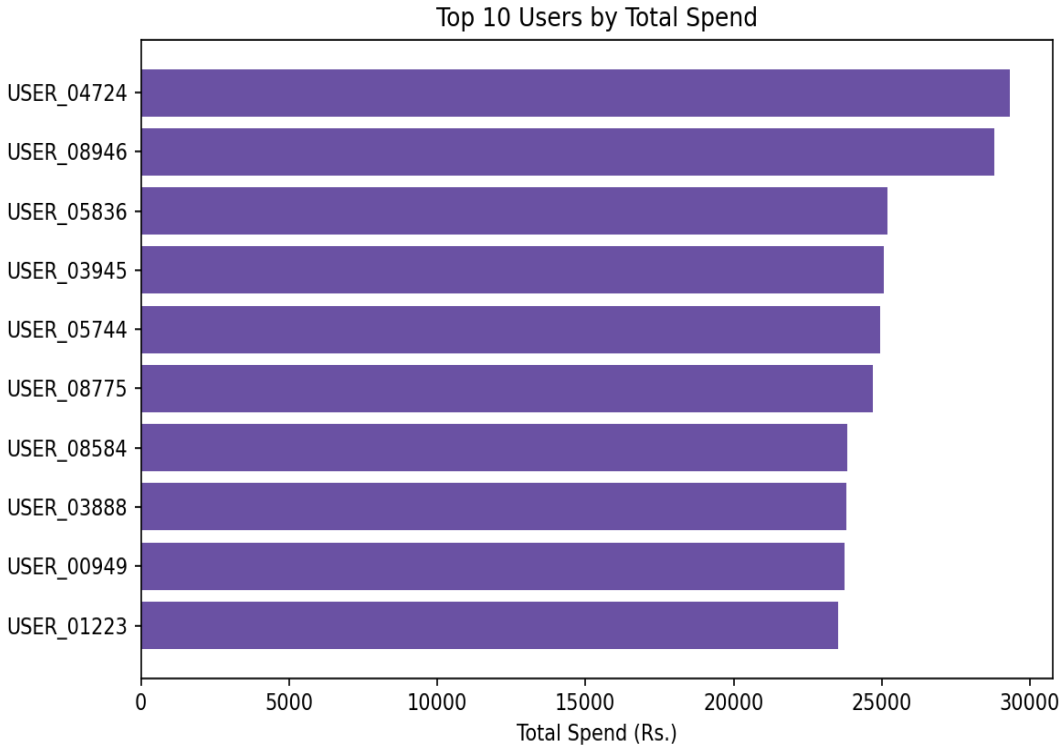


Figure 13. Top 10 users by total spend

INSIGHTS:

- USER_04724 tops the list with a total recorded spend of Rs. 29,315.93 in the supplied data.
- Most users in the dataset transact only a handful of times, so the top-spend list is concentrated among a small group of high-frequency users.
- User-level spend ranking supports loyalty-tier design and targeted retention offers.

KPI:

Top user by spend: USER_04724

Total spend: Rs. 29,315.93

Unique users in dataset: 3,932

16. Average Transaction Value vs. Transaction Volume by Category

The relationship between a category's transaction volume and its average transaction value provides an additional descriptive view of spend composition. The scatter plot should be read as an association rather than proof of causation.

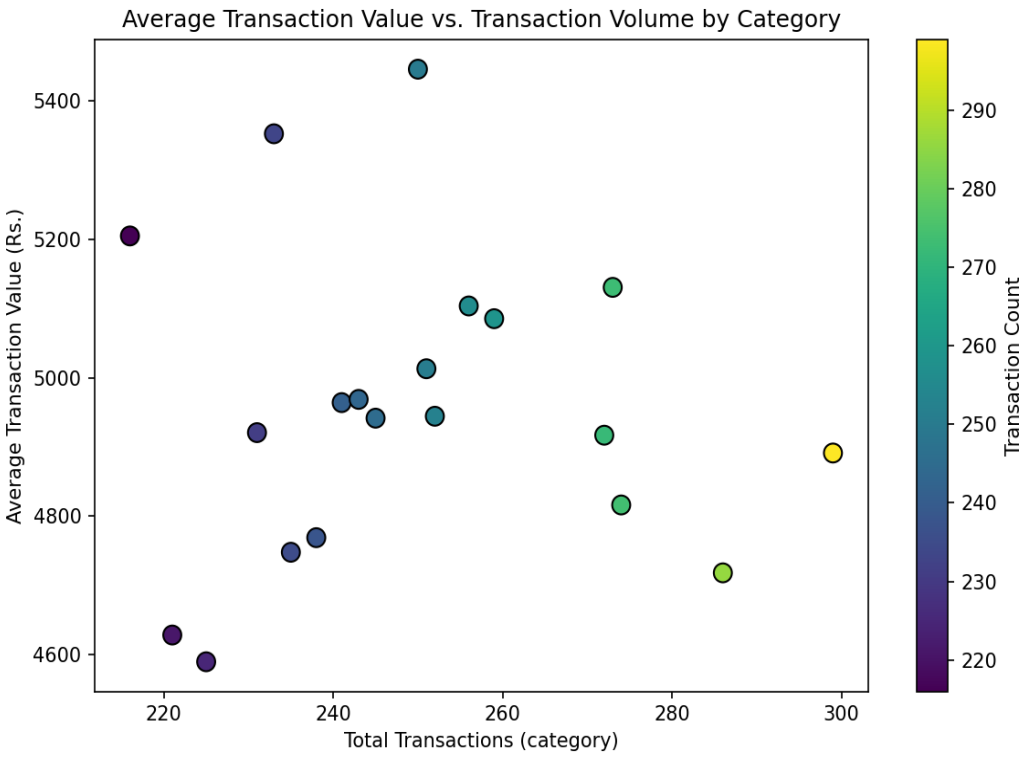


Figure 14. Average transaction value vs. transaction volume by category

INSIGHTS:

- High-frequency categories such as bill payments and mobile recharge tend to carry lower average ticket sizes.
- Gas Bill shows the highest average transaction value among categories in the supplied data, at Rs. 5,446.08.
- This view can support content-gap and cross-sell analysis — pairing high-frequency, low-value categories with higher-value offers.

KPI:

Categories plotted: 20

Highest average-value category: Gas Bill

Its average transaction value: Rs. 5,446.08

17. Transactions by Day of Week

Breaking transaction volume down by day of the week reveals whether wallet activity concentrates on particular days, which can be relevant for campaign scheduling and capacity planning.

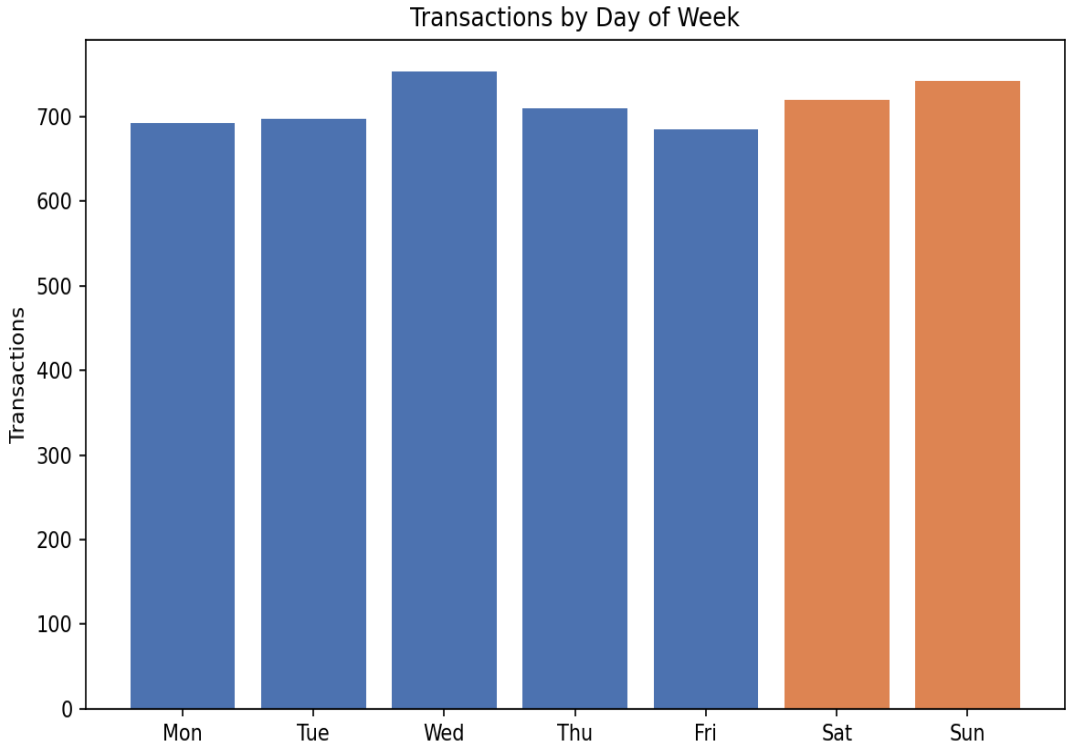


Figure 15. Transactions by day of week

INSIGHTS:

- Wednesday stands out as the busiest day for transactions, with 753 transactions in the supplied data.
- Weekday and weekend volumes are fairly close, unlike purely leisure-driven platforms.
- This pattern can support scheduling of promotional pushes and customer-support staffing.

KPI:

Busiest day: Wednesday

Transactions on that day: 753

18. Weekday vs Weekend Transaction Pattern

Across the row-level observations, average daily transactions are 13.61 on weekdays versus 13.79 on weekends.

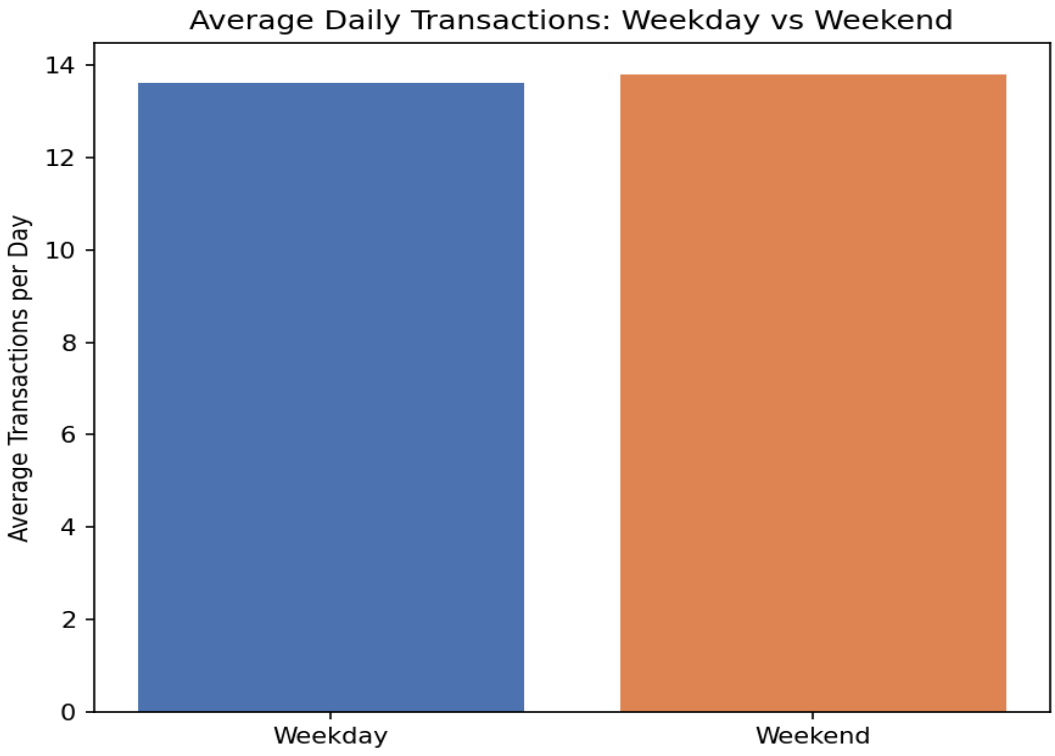


Figure 16. Average daily transactions — weekday vs weekend

INSIGHTS:

- Weekday and weekend averages are close in the supplied data, indicating the wallet is used consistently throughout the week rather than being weekend-driven.
- The dataset is not a complete continuous daily series for every category/merchant combination.
- This result should be treated as a descriptive comparison of the available observations rather than a guaranteed weekly cycle.

KPI:

Avg. transactions per weekday: 13.61

Avg. transactions per weekend day: 13.79

19. Average Transaction Fee & Cashback by Category

Fee and cashback amounts vary by category. Overall, the average transaction fee across the supplied data is Rs. 25.19 and the average cashback is Rs. 50.66. The chart below compares the eight highest-volume categories.

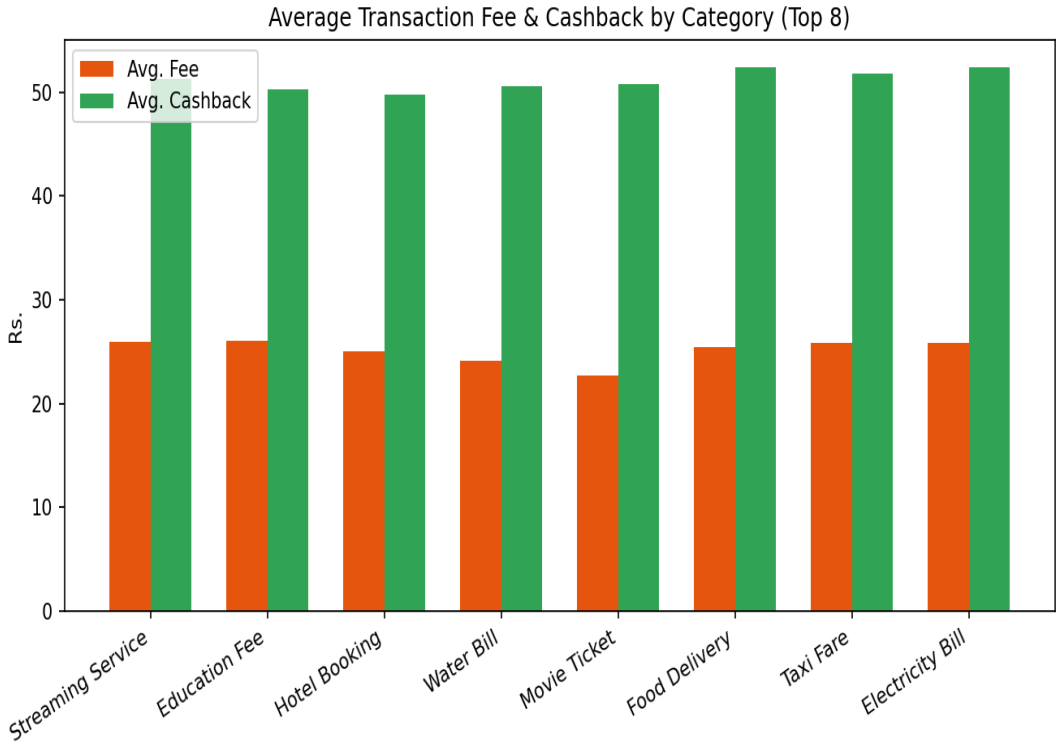


Figure 17. Average transaction fee & cashback by category (top 8)

INSIGHTS:

- Cashback consistently exceeds transaction fees across the highest-volume categories in the supplied data, indicating a net-positive incentive structure for users.
- Fee and cashback levels vary meaningfully by category, which is useful alongside raw volume to understand true category profitability.
- This comparison is a useful catalog-quality indicator alongside transaction counts.

KPI:

Overall average fee: Rs. 25.19

Overall average cashback: Rs. 50.66

20. Transaction Success Rate by Payment Method

Cross-tabulating transaction status against payment method shows whether particular rails are more prone to failure or pending states.

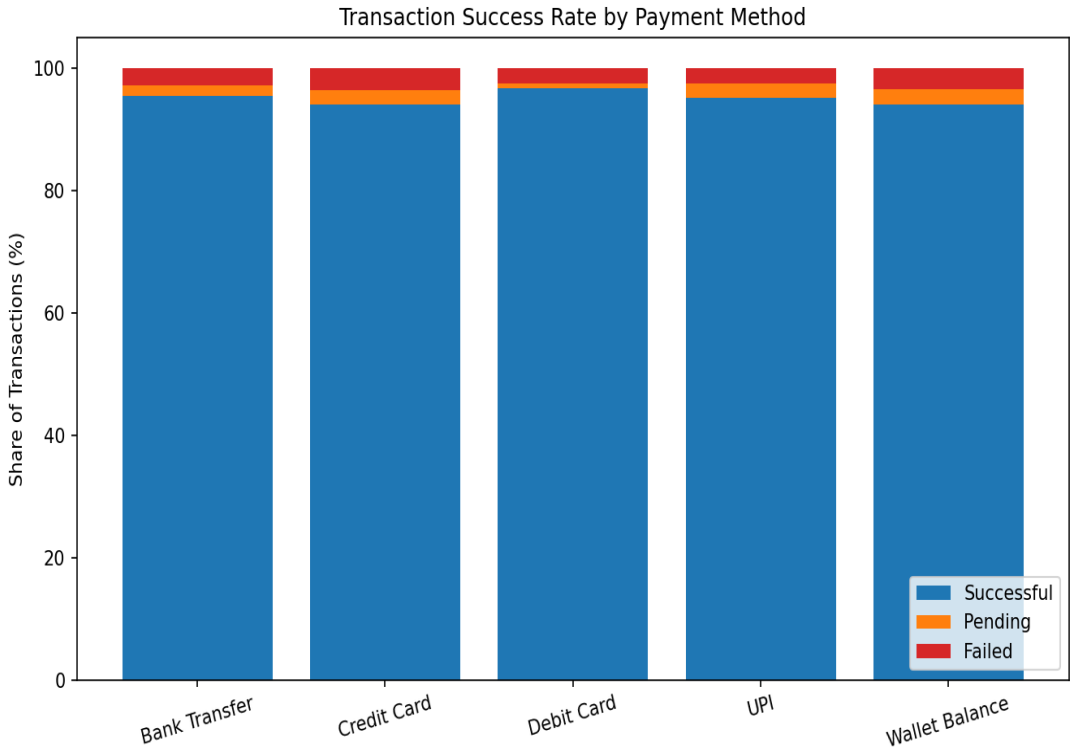


Figure 18. Transaction success rate by payment method

INSIGHTS:

- Debit Card has the highest success rate among payment methods in the supplied data, at 96.67%.
- Credit Card has the lowest success rate at 94.05%, though the gap between methods is modest.
- Success-rate differences by rail can guide retry logic, fallback routing, and risk-team prioritisation.

KPI:

Highest success-rate method: Debit Card (96.67%)

Lowest success-rate method: Credit Card (94.05%)

21. Summary of Key Metrics

Metric	Value
Total transactions	5,000
Successful / Failed / Pending	4,755 / 146 / 99
Date range (transaction date)	2023-08-19 to 2024-08-18
Unique users	3,932
Unique merchants	97
Product categories	20
Top category	Streaming Service (299 transactions)

Top merchant (by transactions)	Airbnb (106 transactions)
Top merchant (by revenue)	Airbnb (Rs. 5,50,248.40)
Most common payment method	Bank Transfer (20.9%)
Average transaction amount	Rs. 4,957.50
Average transaction fee	Rs. 25.19
Average cashback	Rs. 50.66
Average loyalty points	498.79
Total transaction value	Rs. 2,47,87,513.61

22. Next Steps to Move from Descriptive → Predictive

- Demand forecasting by category and merchant to anticipate future transaction-volume needs.
- Anomaly and fraud detection for sudden shifts in transaction rate, category mix, or failure rate.
- Churn / engagement-impact analysis, linking transaction frequency to user retention (requires longer-horizon behavioural data than is present here).
- Category-gap analysis using the value-vs-volume patterns to find under-served, high-value category combinations.
- Trend and cohort analysis once richer, longer-running transaction snapshots become available.

This report answers what the supplied transaction-log snapshot shows. The next analytical layer would use user-level behavioural or settlement-level data, alongside a longer historical window, to explain why transaction patterns emerged and to forecast what is likely to happen next.

23. Glossary

I. Transaction ID — the unique identifier assigned to each wallet transaction in the "transaction_id" field.

II. Product Category — the category label describing what the transaction was for, e.g. "Rent Payment" or "Streaming Service".

III. Payment Method — the funding rail used to complete a transaction (UPI, Debit Card, Credit Card, Bank Transfer, or Wallet Balance).

IV. Transaction Status — whether the transaction completed (Successful), did not complete (Failed), or is still processing (Pending).

V. Loyalty Points — reward points credited to the user for a given transaction, as recorded in the "loyalty_points" field.